Scientific Computing using Python

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mini-course webpage: https://swaprava.wordpress.com/a-short-course-on-python/

Disclaimer: the contents of this lecture series are based on several texts and online resources

Outline of the Talk

1. Part 1: Preliminaries of Python
2. Part 2: Scientific Libraries
3. Part 3: Object Oriented Programming

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- Academic usage: companion of many courses – for 101 CS courses or for supportive computing

Relative popularity of Python

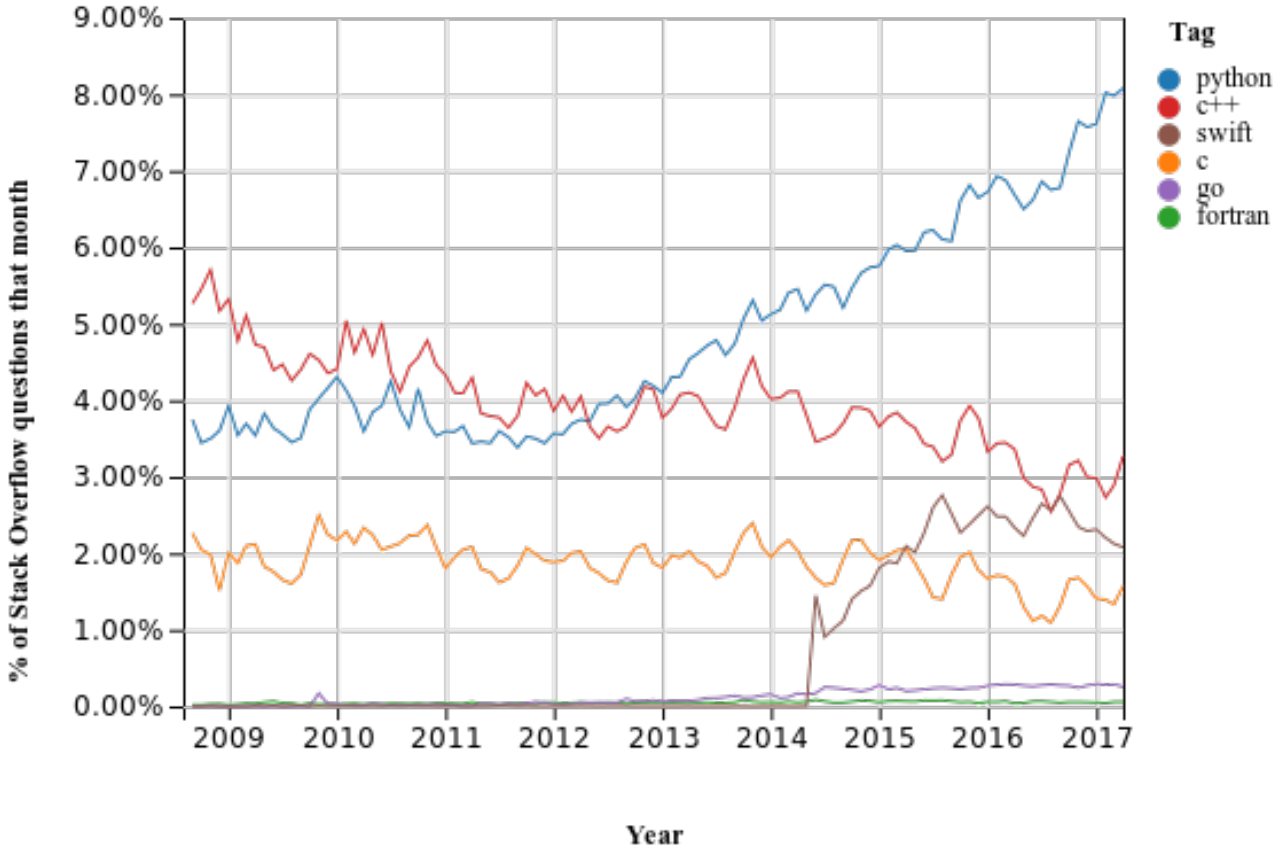

Image courtesy: www.quantecon.org

The plot, produced using Stack Overflow Trends, shows one measure of the relative popularity of Python

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- Usually implemented as interpreted as opposed to compiled (e.g. in C)
- Syntax and design of a python code makes it easier to read, debug, and develop

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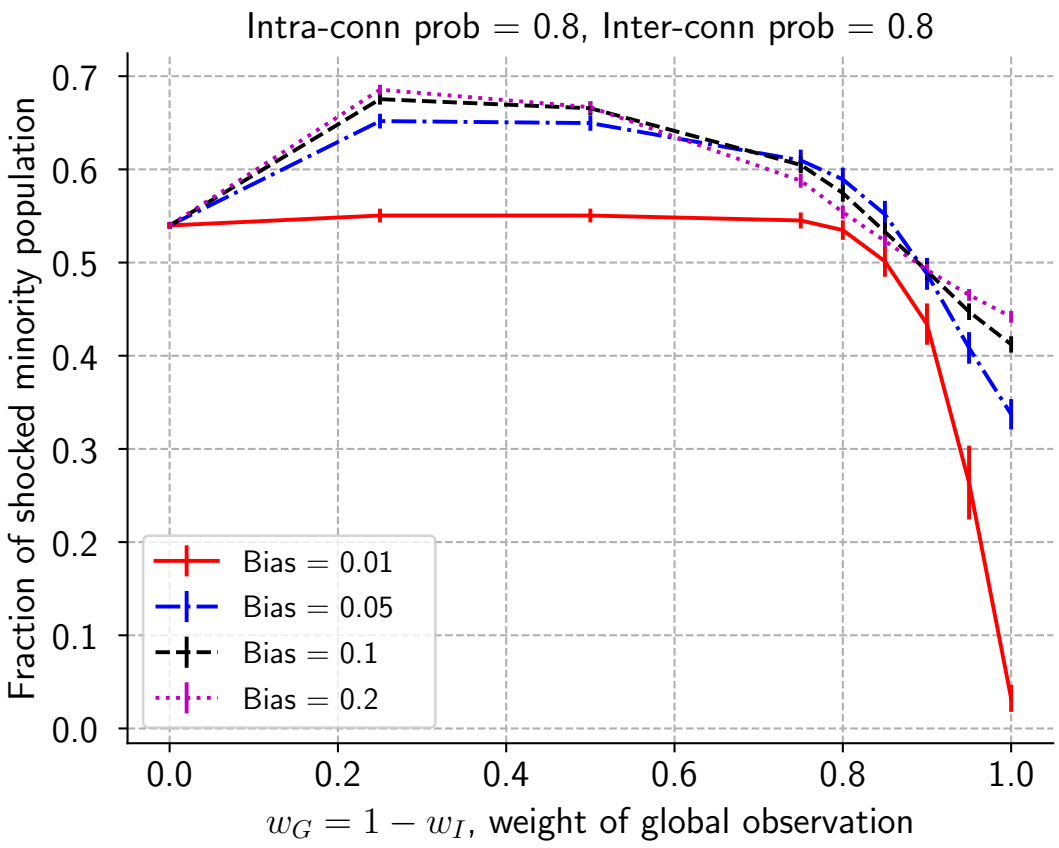
- used in data science, machine learning, artificial intelligence, computational biology, computational physics, quatitative economics etc.

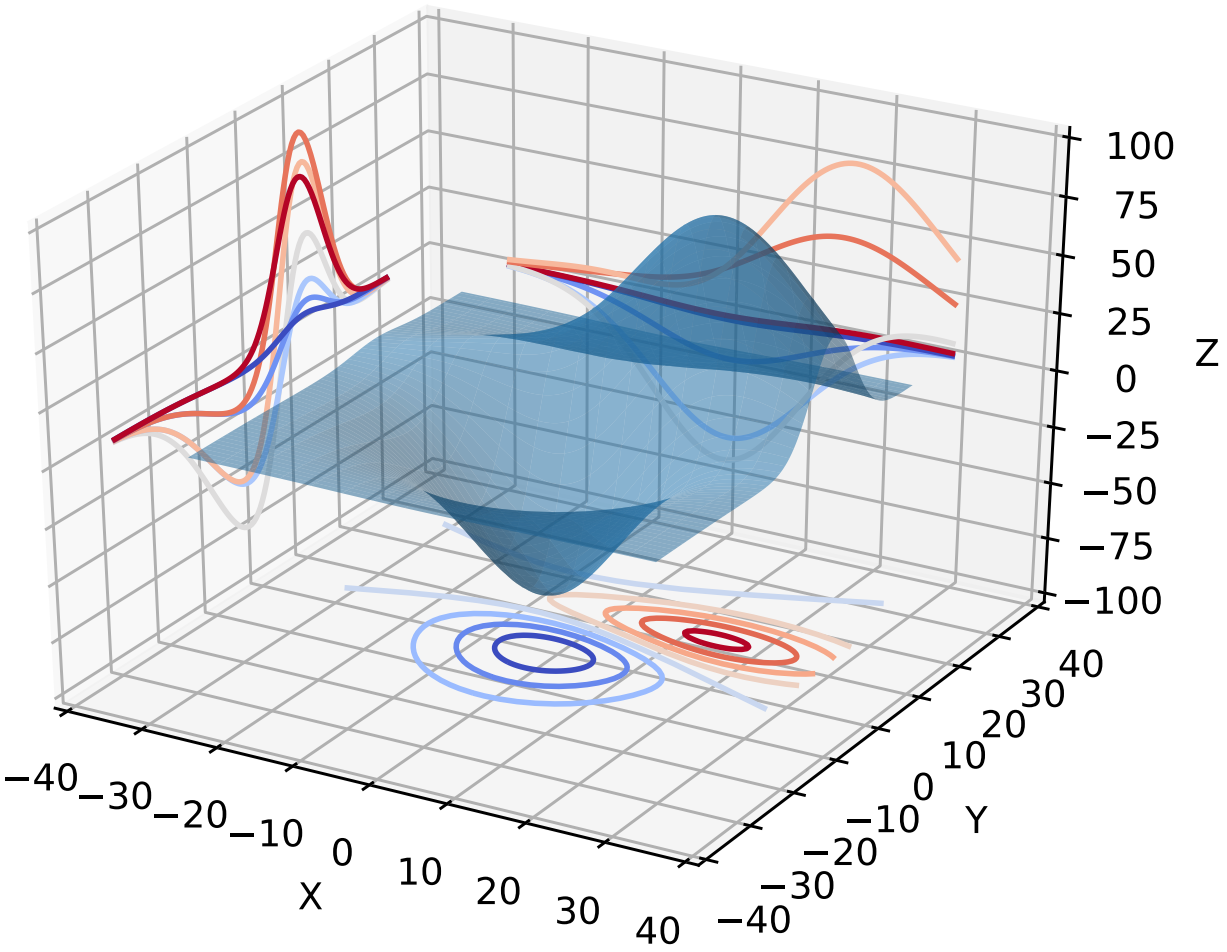
Example in action – 2D plot

- A plot of surprise in the Brexit election



- Used `pandas`, `numpy`, `matplotlib`

Example in action – 3D plot



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- `spyder` is useful for a long codebase development
- Examples

Some Introductory Python Programs

- Task 1: finding if a number is even/odd – `if-else` clause
- Task 2: finding the smallest of three numbers
- Task 3: finding if a natural number is prime or not – `while` loop and `for` loop

- Notice the indentation and absence of any braces or brackets
- Makes the code clutter-free and more readable

More Examples of Loops and Conditionals

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